

## Ques. What Is Union & Union All?

- Both UNION and UNION ALL Operator combine rows from result sets into a single result set.

### UNION

- The union operator **combines** the results of two or more select statements by **removing duplicate rows**.
- The columns and the data types must be the same in select statements.

| ID | NAME    |
|----|---------|
| 1  | Ramesh  |
| 2  | Khilan  |
| 3  | kaushik |

| ID | NAME    |
|----|---------|
| 3  | kaushik |
| 4  | Mohit   |
| 5  | abhay   |

```
Select Column1, Column2, Column3 from Table A
UNION
Select Column1, Column2, Column3 from Table B
```

| ID | NAME    |
|----|---------|
| 1  | Ramesh  |
| 2  | Khilan  |
| 3  | kaushik |
| 4  | Mohit   |
| 5  | abhay   |

## UNION ALL

- The UNION operator selects only distinct values by default. To **allow duplicate values**, use UNION ALL

```
Select Column1, Column2, Column3 from Table A
UNION ALL
Select Column1, Column2, Column3 from Table B
```

```
+----+-----+
| ID | NAME      |
+----+-----+
| 1  | Ramesh    |
| 2  | Khilan    |
| 3  | kaushik   |
| 3  | kaushik   |
| 4  | Mohit     |
| 5  | abhay     |
+----+-----+
```

### Ques. Difference between Union & Union All?

| Union                                                     | Union All                                     |
|-----------------------------------------------------------|-----------------------------------------------|
| Union removes duplicate rows.                             | Union All does not remove the duplicate rows. |
| Union uses a distinct sort                                | Union All does not use a distinct sort        |
| Union can't work with a column that has a text data type. | Union All can work with all data type column. |

## What is MINUS?

- MINUS operator will return only those rows which are **unique(distinct)** in only first SELECT query and not those rows which are **common to both first and second** SELECT queries.

```
-- Employees
| EmpID | EmpName | Department |
| -----|-----|-----|
| 1      | Alice   | HR         |
| 2      | Bob     | IT         |
| 3      | Charlie | Finance    |
| 4      | David   | IT         |
| 5      | Eve     | Marketing  |

-- Managers
| EmpID | EmpName | Department |
| -----|-----|-----|
| 2      | Bob     | IT         |
| 4      | David   | IT         |
| 6      | Frank   | Sales      |
| 7      | Grace   | Marketing  |

-- output:-
SELECT EmpID, EmpName, Department FROM Employees
MINUS
SELECT EmpID, EmpName, Department FROM Managers;
| EmpID | EmpName | Department |
| -----|-----|-----|
| 1      | Alice   | HR         |
| 3      | Charlie | Finance    |
| 5      | Eve     | Marketing  |
```

### Key characteristics of the MINUS operator:

- Returns distinct rows:** Only unique rows from the first query that are not found in the second are returned.
- Requires compatible SELECT statements:** Both SELECT statements involved in the MINUS operation must have the same number of columns, and the corresponding columns must have compatible data types and be in the same order.
- Note:-** While MINUS is a standard SQL operator supported by many database systems (like Oracle, PostgreSQL), **MySQL does not** directly **support** the MINUS operator.

### Achieving the MINUS functionality in MySQL:

- We can achieve the same results as MINUS in MySQL using various techniques, most commonly by combining **LEFT JOIN** and **WHERE** clauses, or using **subqueries** with **NOT IN** or **NOT EXISTS**.

#### 1. Using LEFT JOIN

```
SELECT a.*
FROM table_a AS a
LEFT JOIN table_b AS b ON a.id = b.id
WHERE b.id IS NULL;
```

## 2. Using NOT IN

```
SELECT *  
FROM table_a  
WHERE id NOT IN (SELECT id FROM table_b);
```

## 3. Using NOT EXISTS

```
SELECT *  
FROM table_a AS a  
WHERE NOT EXISTS (SELECT 1 FROM table_b AS b WHERE a.id = b.id);
```

table\_a

| id | value  |
|----|--------|
| 1  | Apple  |
| 2  | Banana |
| 3  | Cherry |
| 4  | Date   |

-----

table\_b

| id | value  |
|----|--------|
| 2  | Banana |
| 4  | Date   |

-----Output:-----

| id | value  |
|----|--------|
| 1  | Apple  |
| 3  | Cherry |

## What is EXCEPT?

- same as minus nothing different.
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